Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/18/2017
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Increasing Cognition Administration Order Flexibility		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	basner@pennmedicine.upenn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-573-5866
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Psychiatry, Division of Sleep and Chronobiology		
PI Address 2:	423 Service Dr, 1013 Blockley Hall		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/01/2016	**End Date:**	04/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NBPF00012		
Performance Goal No.:			
Performance Goal Text:			

Cognition consists of 15 unique versions of its cognitive tests. Currently, these tests are always administered in the same order (i.e., version 1, version 2, …, version 15). This was also the case in the normative study in 19 astronauts, astronaut candidates, and mission controllers performed at Johnson Space Center (JSC). The idea behind keeping the same order for everybody was that, this way, both practice effects and stimulus set difficulty are reflected in the normative data. However, there are two problems with this approach: (1) The strict administration order 1-15 is in practice often broken. Whenever there is a technical or logistical problem that forces a user to switch the testing platform (i.e., use a different laptop), the software often starts with version 1 again. This happened multiple times in various environments including Human Exploration Research. Analog (HERA), Hawaii Space Exploration Analog and Simulation (HI-SEAS), but also International Space Station (ISS) and the TWINS study. (2) Behavior Health and Performance (BHP) Ops seeks greater flexibility in the administration of Cognition, including only administering a subset of tests. These situations are even

Task Description:	more prone to produce test sequences that are out of order and differ between individual tests. At the same time, it is unclear how test administration rate affects practice effects. Therefore, we proposed a study that - allows for disentangling practice effects and stimulus set difficulty effects to make Cognition administration more flexible, and that - investigates the effects of administration rate on practice effects. The study had the following features: (1) We investigated 30 high-functioning subjects (graduate students at Penn or similar) that perform all 15 versions of the 10 tests of the Cognition test battery. (2) 15 subjects performed the battery at an interval of less than or = to 5 days between consecutive tests, while the other half performed the battery at an interval of less than or = to 10 days between consecutive tests. This informs any differences in practice effects depending on administration intervals. (3) The test version administration order was random but balanced (i.e., each version appeared in each position exactly once). (4) Two standard measures of social functioning (Social Skills Inventory, Interpersonal Reactivity Index) were administered during the familiarization session to investigate how the Emotion Recognition Test (ERT) relates to non-cognitive measures of social functioning. This study was augmented (through direct NASA funding) with a third group of 15 subjects that performed Cognition 4x on a single day (with 15 min breaks between tests) on 4 different occasions, mimicking the administration schedule of several past Cognition studies. The major findings of the study are: - Statistically significant practice effects were found for response speed for all tests but the PVT (Psychomotor Vigilance Test) and for accuracy for VOLT (Visual Object Learning Task), F2B (Fractal 2-Back), AM (Abstract Matching), and MRT (Matrix Reasoning Test). - Statistically significant stimulus set effects were found for response speed for VOLT, ERT, MRT, and BART (Balloon Analog Risk Task) and for accuracy for all tests that are not randomly generated. - Administration interval had a significant effect on 2 out of 20 outcomes only (MP speed and DSST (Digit Symbol Substitution Test) accuracy)). Regression estimates were derived that can now be used to adjust for practice effects and stimulus set difficulty effects. In rank correlations (N=30 subjects of NSBRI study) between accuracy and speed on the Emotion Recognition Test (ERT) and 6 subscales of the Social Skills Inventory (SSI) and 4 subscales of the Interpersonal Reactivity Index (IRI), only the fantasy scale of the SSI was statistically significantly correlated with ERT speed (r=-0.402, p=0.0275). Also, the tendency to score SAD more often correlated with a lower overall SSI score (r-0.366, p=0.0469).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project will have substantial impact on progress in three major areas relevant to the needs of NASA and state of the knowledge. 1. Cognition markedly enhances astronauts' and flight physicians' ability to quickly (real-time) and objectively evaluate the neurocognitive status of astronauts relative to activities that can induce fatigue in space (i.e., acute sleep loss from prolonged duty, chronic sleep restriction, inadequate recovery sleep, slam shifts and circadian misalignment, high physical and/or cognitive workloads, EVAs); relative to fatigue countermeasures, e.g., different sleep-wake schedules, sleep-promoting and wake-promoting medications, light exposure for circadian associated with occult neurobehavioral risks in space (e.g., space fog, space asthenia/neurasthenia). 2. Cognition permits identification of important fatigue-related individual differences (i.e., differential vulnerability) in the nature and severity of cognitive performance deficits (e.g., from deficits in spatial orientation, to working memory, to abstract reasoning, to risk decision-making) during space flight, in a comprehensive and precise manner to permit optimal targeting of fatigue countermeasures to specific individuals, and to help predict the performance capability of individual astronauts relative to specific space flight tasks (i.e., align cognitive performance readiness relative to the need to conduct specific space flight tasks). 3. Cognition helps in the medical identification and treatment management course of neurologically-based performance deficits in space flight due to environmental stressors (e.g., exposure to high CO2, hypoxia, radiation); medically urgent events (e.g., head injury, papilledema and/or the possibility of elevated intracranial pressure [ICP]); and neurobehavioral conditions brought on by prolonged stays in space (e.g., time in confinement, neural remodeling from sensorimotor alterations, affective disorders). Although the Cognition test battery is primarily developed for space flight, it will be a valuable tool in many Earth-based patient and non-patient population settings, where identification of suboptimal cognitive performance is important for safe operations (e.g., truck drivers, operators of heavy machinery) or for tracking therapeutic effectiveness. Cognition is optimized for repeated administration, a feature that many of the currently available test batteries are lacking.
Task Progress:	All 30 subjects were recruited and data acquisition was completed in all but one subject. The latter dropped out after 6 administrations, but was replaced with another subject. As of 5/31/2017, data acquisition was complete.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Basner M, Hermosillo E, Nasrini J, McGuire S, Saxena S, Moore TM, Gur RC, Dinges DF. "Repeated administration effects on Psychomotor Vigilance Test (PVT) performance." Sleep. 2018 Jan 1;41(1). https://doi.org/10.1093/sleep/zsx187 ; PubMed PMID: 29126328 , Jan-2018
Articles in Peer-reviewed Journals	Lee G, Moore TM, Basner M, Nasrini J, Roalf DR, Ruparel K, Port AM, Dinges DF, Gur RC. "Age, sex, and repeated measures effects on NASA's "Cognition" test battery in STEM educated adults." Aerosp Med Hum Perform. 2020 Jan 1;91(1):18-25. https://doi.org/10.3357/AMHP.5485.2020 ; PubMed PMID: 31852569 , Jan-2020

Articles in Peer-reviewed Journals	Basner M, Hermosillo E, Nasrini J, Saxena S, Dinges DF, Moore TM, Gur RC. "Cognition test battery: Adjusting for practice and stimulus set effects for varying administration intervals in high performing individuals." J Clin Exp Neuropsychol. 2020 Jul;42(5):516-29. https://doi.org/10.1080/13803395.2020.1773765 ; PMID: 32539487; PMCID: PMC7375457 [Erratum in: J Clin Exp Neuropsychol. 2020 Jun 24;:1.] , Jul-2020
Articles in Peer-reviewed Journals	Basner M, Moore TM, Nasrini J, Gur RC, Dinges DF. "Response speed measurements on the Psychomotor Vigilance Test: How precise is precise enough?" Sleep. 2021 Jan 21;44(1):zsaa121. Epub 2020 Jul 2. https://doi.org/10.1093/sleep/zsaa121 ; PMID: 32556295 , Jan-2021
Articles in Peer-reviewed Journals	Basner M, Moore TM, Hermosillo E, Nasrini J, Dinges DF, Gur RC, Johannes B. "Cognition test battery performance is associated with simulated 6df spacecraft docking performance." Aerosp Med Hum Perform. 2020 Nov;91(11):861-7. https://doi.org/10.3357/AMHP.5602.2020 ; PMID: 33334406; PMCID: PMC7755107 , Nov-2020
Articles in Peer-reviewed Journals	Smith MG, Wusk GC, Nasrini J, Baskin P, Dinges DF, Roma PG, Basner M. "Effects of six weeks of chronic sleep restriction with weekend recovery on cognitive performance and wellbeing in high-performing adults." Sleep. 2021 Aug 13;44(8):zsab051. https://doi.org/10.1093/sleep/zsab051 ; PMID: 33630069 , Aug-2021